

Institutions Required for both Real and Imaginary Arbitrage in the Cyclotomic Economy with 9 Districts

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Abstract

This paper studies the institutional foundations of real and imaginary arbitrage in a nine-district cyclotomic economy. The economy is governed by the ninth cyclotomic polynomial

$$\Phi_9(x) = x^6 + x^3 + 1,$$

with cyclic district operator P satisfying $P^9 = I$. District-level economic quantities are complex-valued: their real components represent material, settled, or realized magnitudes, while their imaginary components represent expectations, shadow values, informational premia, liquidity beliefs, ambiguity, and strategic anticipation. The paper identifies the institutional stack required to support real arbitrage and imaginary arbitrage: property rights, contract enforcement, settlement systems, logistics, clearinghouses, exchanges, banks, disclosure rules, statistical agencies, auditing, derivative markets, insurance, communication systems, and credible public authorities. It proves that real arbitrage compresses material price and return dispersion, while imaginary arbitrage compresses expectation and shadow-price dispersion. In the sign-state version of the model, four complex variables—scarcity, price, rate, and risk premium—generate a raw bound of $3^8 = 6561$ possible sign-distinguishable district types. Strong arbitrage of both real and imaginary financial components reduces the effective bound to $3^2 = 9$ if complex scarcity remains district-specific, and to 1 if complex scarcity is also equalized. The framework connects mathematics, economics, finance, political science, geopolitics, international relations, warfare economics, statistics, algorithms, and AI/ML under a unified institutional theory of cyclotomic arbitrage.

The paper ends with “The End”

Keywords: cyclotomic economics, cyclotomic economic fields, nine-district model, institutions.

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1 Introduction

A cyclotomic economy is an economy whose regional, financial, and institutional structure is governed by a cyclotomic polynomial. In the nine-district case, the relevant polynomial is

$$\Phi_9(x) = x^6 + x^3 + 1.$$

Let the districts be indexed by

$$i \in \mathbb{Z}_9 = \{0, 1, 2, \dots, 8\}.$$

Let P denote the cyclic shift operator:

$$(Px)_i = x_{i-1},$$

with indices taken modulo 9. Since $P^9 = I$, the cyclotomic operator is

$$\Phi_9(P) = I + P^3 + P^6.$$

The earlier paper *Cyclotomic Economic Field Theory with Nine Districts* introduced the core nine-district economy and the primitive clearing condition

$$\Phi_9(P)z = 0,$$

where $z \in \mathbb{R}^9$ is the vector of district net surpluses [1]. Equivalently,

$$z_i + z_{i+3} + z_{i+6} = 0, \quad i = 0, 1, 2.$$

This paper extends that framework by studying the institutions required for both real arbitrage and imaginary arbitrage.

The central idea is that a complex economic quantity

$$X_i = X_i^R + iX_i^I$$

has two economically distinct parts. The real part X_i^R represents material or settled reality: actual prices, realized returns, physical scarcity, and enforceable claims. The imaginary part X_i^I represents anticipatory or shadow reality: expected scarcity, liquidity anxiety, ambiguity, model risk, geopolitical fear, and future-state valuation.

Real arbitrage compresses the real components. Imaginary arbitrage compresses the imaginary components. Real arbitrage requires institutions that move goods, money, collateral, and claims. Imaginary arbitrage requires institutions that move information, forecasts, beliefs, contingent claims, and credible signals.

2 The Nine-District Cyclotomic Economy

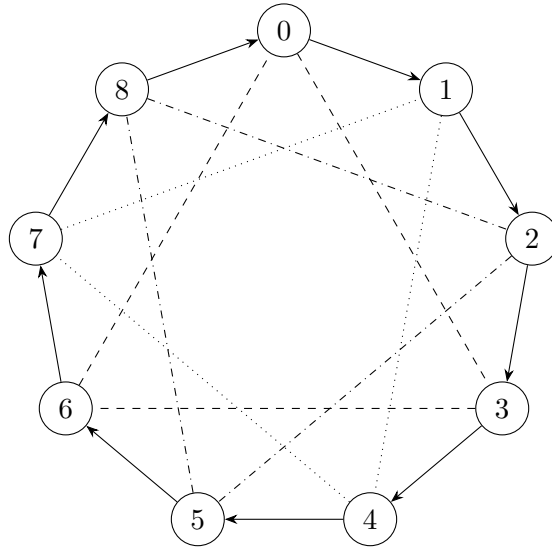
2.1 District Structure

The nine districts are placed on a cycle:

$$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 8 \rightarrow 0.$$

The hidden triads are

$$\mathcal{T}_0 = \{0, 3, 6\}, \quad \mathcal{T}_1 = \{1, 4, 7\}, \quad \mathcal{T}_2 = \{2, 5, 8\}.$$



nine-district cycle with hidden cyclotomic triads

Figure 1: The nine-district cyclotomic economy.

2.2 Primitive Field

Definition 1 (Primitive Cyclotomic Field). *The primitive cyclotomic field is*

$$\mathcal{H}_9 = \text{Ker } \Phi_9(P) = \{x \in \mathbb{R}^9 : x_i + x_{i+3} + x_{i+6} = 0, i = 0, 1, 2\}.$$

The projection onto the primitive field is

$$\Pi_9 = I - \frac{1}{3}(I + P^3 + P^6).$$

For $x \in \mathbb{R}^9$,

$$(\Pi_9 x)_i = x_i - \frac{x_i + x_{i+3} + x_{i+6}}{3}.$$

Proposition 1 (Dimension of the Primitive Field). *The primitive field $\mathcal{H}_9$ has dimension 6.*

Proof. The condition $x_i + x_{i+3} + x_{i+6} = 0$ gives one independent linear restriction for each residue class modulo 3. Therefore there are three independent restrictions in $\mathbb{R}^9$. Hence

$$\dim \mathcal{H}_9 = 9 - 3 = 6.$$

Equivalently, the primitive ninth-root exponents are

$$\{1, 2, 4, 5, 7, 8\},$$

which also gives six primitive Fourier modes. □

3 Complex Economic Quantities

Each district i has four complex state variables:

$$\begin{aligned} \mathfrak{s}_i &= s_i^R + i s_i^I, \\ \pi_i &= p_i^R + i p_i^I, \\ \rho_i &= r_i^R + i r_i^I, \\ \mu_i &= \mu_i^R + i \mu_i^I. \end{aligned}$$

These represent complex scarcity, complex price, complex rate, and complex risk premium.

Table 1: Interpretation of complex economic variables

Variable	Real component	Imaginary component
$\mathfrak{s}_i$	material scarcity	expected or strategic scarcity
π_i	spot or settled price	shadow or anticipatory price
ρ_i	realized financing rate	liquidity, duration, or expected-rate component
μ_i	ordinary risk premium	ambiguity, tail-risk, or geopolitical premium

Definition 2 (District Sign State). *The sign state of district i is*

$$\sigma_i = (\text{sgn } s_i^R, \text{sgn } s_i^I, \text{sgn } p_i^R, \text{sgn } p_i^I, \text{sgn } r_i^R, \text{sgn } r_i^I, \text{sgn } \mu_i^R, \text{sgn } \mu_i^I) \in \{-1, 0, +1\}^8.$$

Since there are eight ternary sign coordinates, the raw sign-state count is

$$3^8 = 6561.$$

Proposition 2 (Raw Sign-State Bound). *If districts are distinguished solely by the one-period signs of complex scarcity, price, rate, and risk premium, then*

$$n_{\max}^{\text{raw}} = 3^8 = 6561.$$

Proof. Each of the four complex variables contributes two real sign coordinates. Thus the number of independent sign coordinates is $2 \cdot 4 = 8$. Each coordinate has three possible values, $-1, 0, +1$. Therefore the number of possible sign states is $3^8 = 6561$. □

4 Real and Imaginary Arbitrage

4.1 Real Arbitrage

Real arbitrage equalizes real components of economic variables across connected districts, up to transaction costs.

Let $G^R = (V, E^R)$ be the real-arbitrage network, where

$$V = \mathbb{Z}_9.$$

If $(i, j) \in E^R$, then real arbitrage imposes

$$|X_i^R - X_j^R| \leq \tau_{ij}^R,$$

where $\tau_{ij}^R \geq 0$ is the material transaction-cost wedge.

In the frictionless case,

$$\tau_{ij}^R = 0,$$

so

$$X_i^R = X_j^R.$$

Definition 3 (Real Arbitrage). *A real arbitrage institution is an institutional arrangement that reduces τ_{ij}^R for goods, claims, money, collateral, or realized returns between districts i and j .*

Real arbitrage requires: property rights, contract enforcement, money and payments, settlement, logistics, collateral, clearing.

4.2 Imaginary Arbitrage

Imaginary arbitrage equalizes imaginary components of economic variables across connected districts, up to information and credibility costs.

Let $G^I = (V, E^I)$ be the imaginary-arbitrage network. If $(i, j) \in E^I$, then

$$|X_i^I - X_j^I| \leq \tau_{ij}^I,$$

where $\tau_{ij}^I \geq 0$ is an information, ambiguity, credibility, or model-risk wedge.

In the frictionless informational case,

$$\tau_{ij}^I = 0,$$

so

$$X_i^I = X_j^I.$$

Definition 4 (Imaginary Arbitrage). *An imaginary arbitrage institution is an institutional arrangement that reduces τ_{ij}^I for expectations, shadow prices, beliefs, forecasts, contingent claims, liquidity expectations, or strategic anticipations.*

Imaginary arbitrage requires: statistics, disclosure, auditing, forecasting, communication networks, derivatives, insurance, and credible policy signals.

4.3 Full Complex Arbitrage

Definition 5 (Full Complex Arbitrage). *Full complex arbitrage on an edge (i, j) holds for a complex variable X if both*

$$|X_i^R - X_j^R| \leq \tau_{ij}^R$$

and

$$|X_i^I - X_j^I| \leq \tau_{ij}^I.$$

In the frictionless limit,

$$X_i = X_j.$$

5 Institutional Stack

Table 2: Institutional requirements for real and imaginary arbitrage

Institution	Real-arbitrage function	Imaginary-arbitrage function
Property rights	Defines ownership of goods and claims	Defines ownership of contingent claims
Contract law	Enforces delivery and payment	Enforces future-state contracts
Courts and arbitration	Resolves disputes over real obligations	Resolves disputes over contingent interpretations
Payment system	Moves money across districts	Supports rapid repricing of beliefs
Clearinghouse	Nets obligations and reduces settlement risk	Nets derivative and expectation-linked claims
Banks and brokers	Finance arbitrage positions	Intermediate expectation-sensitive claims
Exchanges and dealers	Provide price discovery	Provide shadow-price discovery
Custody and collateral systems	Secure assets and collateral	Validate pledgeable future-state claims
Transport and logistics	Move goods and inventories	Reveal bottlenecks and expected shortages
Disclosure rules	Reduce hidden material information	Reduce asymmetric expectations
Auditing and accounting	Standardize balance sheets	Validate risk and liquidity narratives
Statistical agencies	Measure quantities and prices	Anchor shared macro expectations
Derivatives and insurance	Hedge real exposures	Price future, tail, and ambiguity states
Communication networks	Coordinate trade	Coordinate beliefs and forecasts
Central bank or regulator	Provides liquidity and rule credibility	Anchors expectations and policy reaction functions

The shortest institutional characterization is:

real arbitrage requires enforceable trade,

imaginary arbitrage requires enforceable information-contingent trade.

6 Financial Arbitrage Conditions

Let R_i denote a realized return in district i . A complex asset-pricing relation may be written as

$$\mathbb{E}[R_i] = \rho_i + \mu_i.$$

If liquidity premia λ_i are explicitly included, one may write

$$\mathbb{E}[R_i] = \rho_i + \mu_i + \lambda_i.$$

Here, this paper keeps the four-variable scarcity-price-rate-risk-premium system, while recognizing that liquidity can be embedded in the imaginary component of rates or risk premia.

A risk-adjusted arbitrage condition between districts i and j is

$$\mathbb{E}[R_i] - \rho_i - \mu_i = \mathbb{E}[R_j] - \rho_j - \mu_j.$$

Componentwise:

$$\mathbb{E}[R_i^R] - r_i^R - \mu_i^R = \mathbb{E}[R_j^R] - r_j^R - \mu_j^R,$$

$$\mathbb{E}[R_i^I] - r_i^I - \mu_i^I = \mathbb{E}[R_j^I] - r_j^I - \mu_j^I.$$

Definition 6 (Componentwise Financial Arbitrage). *Componentwise financial arbitrage holds between districts i and j if*

$$p_i^R = p_j^R, \quad p_i^I = p_j^I,$$

$$r_i^R = r_j^R, \quad r_i^I = r_j^I,$$

and

$$\mu_i^R = \mu_j^R, \quad \mu_i^I = \mu_j^I,$$

after admissible transaction-cost and information wedges.

7 Cyclotomic No-Arbitrage

Let $z \in \mathbb{R}^9$ be a feasible surplus vector satisfying

$$\Phi_9(P)z = 0.$$

Let complex prices be

$$\pi = p^R + ip^I.$$

Complex no-arbitrage requires

$$\pi^\top z = 0$$

for every feasible primitive circulation $z \in \mathcal{H}_9$. Equivalently,

$$(p^R)^\top z = 0$$

and

$$(p^I)^\top z = 0.$$

Theorem 1 (Triadic Complex Price Anchors). *If complex no-arbitrage holds for every $z \in \mathcal{H}_9$, then*

$$p_0^R = p_3^R = p_6^R, \quad p_1^R = p_4^R = p_7^R, \quad p_2^R = p_5^R = p_8^R,$$

and

$$p_0^I = p_3^I = p_6^I, \quad p_1^I = p_4^I = p_7^I, \quad p_2^I = p_5^I = p_8^I.$$

Proof. The real part follows by choosing primitive circulation vectors such as $e_0 - e_3$, $e_3 - e_6$, and analogues in the other triads. Since $(p^R)^\top z = 0$ for all $z \in \mathcal{H}_9$, we get equality of real prices inside each triad. The imaginary part follows identically from $(p^I)^\top z = 0$. Therefore both real and imaginary price components are anchored triadically. $\square$

8 Arbitrage Compression of District Types

Let d_{eff} denote the number of independent ternary sign coordinates remaining after institutions impose arbitrage and communication constraints. Then the effective sign-state bound is

$$n_{\text{max}}^{\text{eff}} = 3^{d_{\text{eff}}}.$$

Without institutional compression:

$$d_{\text{eff}} = 8,$$

so

$$n_{\text{max}}^{\text{raw}} = 3^8 = 6561.$$

8.1 Strong Financial Arbitrage

Suppose componentwise arbitrage equalizes price, rate, and risk-premium signs across all connected districts:

$$\begin{aligned} \text{sgn } p_i^R &= \text{sgn } p_j^R, & \text{sgn } p_i^I &= \text{sgn } p_j^I, \\ \text{sgn } r_i^R &= \text{sgn } r_j^R, & \text{sgn } r_i^I &= \text{sgn } r_j^I, \\ \text{sgn } \mu_i^R &= \text{sgn } \mu_j^R, & \text{sgn } \mu_i^I &= \text{sgn } \mu_j^I. \end{aligned}$$

Then the six financial sign coordinates cease to be district-specific. If complex scarcity remains district-specific, only

$$\text{sgn } s_i^R, \quad \text{sgn } s_i^I$$

remain.

Therefore

$$d_{\text{eff}} = 2,$$

and

$$n_{\text{max}}^{\text{eff}} = 3^2 = 9.$$

Theorem 2 (Financial Arbitrage Compression). *In a connected nine-district cyclotomic economy, if componentwise arbitrage equalizes the real and imaginary components of prices, rates, and risk premia, while complex scarcity remains district-specific, then the effective sign-state bound is*

$$n_{\text{max}}^{\text{eff}} = 9.$$

Proof. There are initially eight sign coordinates. Componentwise arbitrage equalizes the six signs associated with π_i, ρ_i, μ_i . These six signs become common across districts and no longer distinguish districts. The two scarcity signs remain district-specific. Hence there are $3^2 = 9$ possible district-specific sign states. $\square$

8.2 Full Goods, Financial, and Informational Integration

If complex scarcity is also equalized:

$$\text{sgn } s_i^R = \text{sgn } s_j^R, \quad \text{sgn } s_i^I = \text{sgn } s_j^I,$$

then no sign coordinate remains district-specific:

$$d_{\text{eff}} = 0.$$

Thus

$$n_{\text{max}}^{\text{eff}} = 1.$$

Corollary 1 (Complete Institutional Compression). *If real arbitrage, imaginary arbitrage, goods arbitrage, and communication equalize all eight sign coordinates, then*

$$n_{\text{max}}^{\text{eff}} = 1.$$

9 Political Economy of Arbitrage Institutions

Arbitrage institutions are not neutral. They allocate power across districts by determining which differences can survive.

Let A^R be the adjacency matrix of the real-arbitrage network and A^I the adjacency matrix of the imaginary-arbitrage network. Define institutional intensities

$$\alpha_{ij}^R \in [0, 1], \quad \alpha_{ij}^I \in [0, 1],$$

where higher values mean stronger arbitrage institutions.

Let real and imaginary wedges be

$$\tau_{ij}^R = \bar{\tau}_{ij}^R(1 - \alpha_{ij}^R),$$

$$\tau_{ij}^I = \bar{\tau}_{ij}^I(1 - \alpha_{ij}^I).$$

Political authorities choose institutional intensity subject to a budget:

$$\sum_{(i,j)} c_{ij}^R \alpha_{ij}^R + \sum_{(i,j)} c_{ij}^I \alpha_{ij}^I \leq B.$$

A political planner may maximize

$$W = \sum_{i=0}^8 u_i - \frac{\kappa_R}{2} \sum_{(i,j)} (|X_i^R - X_j^R| - \tau_{ij}^R)_+^2 - \frac{\kappa_I}{2} \sum_{(i,j)} (|X_i^I - X_j^I| - \tau_{ij}^I)_+^2.$$

This formulation captures a central political-economy tradeoff: stronger institutions reduce arbitrage wedges but also reduce local autonomy, local rents, and district-specific strategic leverage.

10 Geopolitics and International Relations

External actors can attack or support arbitrage institutions. Let

$$h^R \in \mathbb{R}^9$$

be a real institutional pressure vector, representing sanctions, blockades, capital controls, infrastructure attacks, or trade restrictions. Let

$$h^I \in \mathbb{R}^9$$

be an imaginary institutional pressure vector, representing propaganda, uncertainty, credibility attacks, misinformation, expectation shocks, or diplomatic signaling.

The primitive geopolitical components are

$$h^{R,\text{prim}} = \Pi_9 h^R,$$

$$h^{I,\text{prim}} = \Pi_9 h^I.$$

A geopolitical attack is primitive if

$$h^R = \Pi_9 h^R \quad \text{or} \quad h^I = \Pi_9 h^I.$$

Such an attack may leave aggregate indicators unchanged while disrupting primitive district relations.

Proposition 3 (Hidden Geopolitical Disruption). *If $h^I \in \mathcal{H}_9$, then an external actor can disrupt imaginary arbitrage without changing the aggregate imaginary pressure.*

Proof. If $h^I \in \mathcal{H}_9$, then it has zero triadic means and no aggregate component. Therefore aggregate measurement may fail to detect it, even though it affects primitive district-level expectation structure. $\square$

11 Warfare Economics and Security

Warfare and coercion affect arbitrage by raising real and imaginary wedges. Let m_i denote district security capacity and $\bar{m}_i$ rival pressure. A contest-success function is

$$q_i = \frac{m_i^\eta}{m_i^\eta + \bar{m}_i^\eta}, \quad \eta > 0.$$

Real disruption is

$$\omega_i^R = \bar{\omega}_i^R(1 - q_i) + \xi_i^R,$$

and imaginary disruption is

$$\omega_i^I = \bar{\omega}_i^I(1 - q_i) + \xi_i^I.$$

The real arbitrage wedge becomes

$$\tau_{ij}^R = \tau_{ij,0}^R + \gamma_R(\omega_i^R + \omega_j^R),$$

while the imaginary arbitrage wedge becomes

$$\tau_{ij}^I = \tau_{ij,0}^I + \gamma_I(\omega_i^I + \omega_j^I).$$

Thus conflict does not merely destroy output. It destroys arbitrage institutions by increasing the cost of moving goods, enforcing contracts, trusting data, believing commitments, and pricing future states.

12 Statistical Measurement

Suppose the econometrician observes complex variables

$$X_i(t) = X_i^R(t) + iX_i^I(t)$$

for $i = 0, \dots, 8$, $t = 1, \dots, T$.

Define real and imaginary arbitrage dispersion:

$$D_R(t) = \sum_{(i,j) \in E^R} (X_i^R(t) - X_j^R(t))^2,$$

$$D_I(t) = \sum_{(i,j) \in E^I} (X_i^I(t) - X_j^I(t))^2.$$

Define cyclotomic primitive dispersion:

$$C_R(t) = \|\Pi_9 X^R(t)\|^2,$$

$$C_I(t) = \|\Pi_9 X^I(t)\|^2.$$

Real arbitrage is statistically strong when $D_R(t)$ is small. Imaginary arbitrage is statistically strong when $D_I(t)$ is small. Primitive arbitrage failure is large when $C_R(t)$ or $C_I(t)$ is large.

A regression specification is

$$D_R(t+1) = a_R - b_R I_R(t) + c_R W_R(t) + \varepsilon_R(t+1),$$

$$D_I(t+1) = a_I - b_I I_I(t) + c_I W_I(t) + \varepsilon_I(t+1),$$

where I_R, I_I are institutional-quality indices and W_R, W_I are real and imaginary conflict wedges. The theory predicts

$$\begin{aligned} b_R &> 0, & b_I &> 0, \\ c_R &> 0, & c_I &> 0. \end{aligned}$$

13 Algorithms

Table 3: Algorithm 1: Detecting real and imaginary arbitrage failure

Step	Operation
1	Input complex district panel $X_i(t) = X_i^R(t) + iX_i^I(t)$.
2	Construct real and imaginary arbitrage networks G^R and G^I .
3	Compute real dispersion $D_R(t) = \sum_{(i,j) \in E^R} (X_i^R(t) - X_j^R(t))^2$.
4	Compute imaginary dispersion $D_I(t) = \sum_{(i,j) \in E^I} (X_i^I(t) - X_j^I(t))^2$.
5	Compute primitive dispersions $C_R(t) = \ \Pi_9 X^R(t)\ ^2$ and $C_I(t) = \ \Pi_9 X^I(t)\ ^2$.
6	Flag real arbitrage failure if $D_R(t)$ or $C_R(t)$ exceeds its threshold.
7	Flag imaginary arbitrage failure if $D_I(t)$ or $C_I(t)$ exceeds its threshold.

Table 4: Algorithm 2: Estimating effective sign-state capacity

Step	Operation
1	Input signs of $\mathfrak{s}_i, \pi_i, \rho_i, \mu_i$ for all districts.
2	Remove sign coordinates equalized by real arbitrage.
3	Remove sign coordinates equalized by imaginary arbitrage.
4	Remove sign coordinates functionally implied by asset-pricing identities.
5	Let d_{eff} be the number of remaining independent ternary coordinates.
6	Return $n_{\text{max}}^{\text{eff}} = 3^{d_{\text{eff}}}$.

14 AI and Machine Learning

A machine-learning model for arbitrage institutions should respect the cyclic symmetry of the district economy. Let $x_t \in \mathbb{R}^9$ represent a district vector. An equivariant linear layer has the form

$$f_\theta(x_t) = \sum_{k=0}^8 \theta_k P^k x_t.$$

A primitive-filtered model is

$$h_{t+1} = \sigma \left(\Pi_9 \sum_{k=0}^8 \theta_k P^k h_t + B u_t \right),$$

where u_t contains institutional variables.

For complex variables, one may separately model real and imaginary components:

$$\hat{X}_{t+1}^R = f_{\theta_R}(X_t^R, u_t),$$

$$\hat{X}_{t+1}^I = f_{\theta_I}(X_t^I, u_t).$$

A cyclotomic institutional loss is

$$\mathcal{L}(\theta) = \sum_t \left\| X_{t+1}^R - \hat{X}_{t+1}^R \right\|^2 + \sum_t \left\| X_{t+1}^I - \hat{X}_{t+1}^I \right\|^2 + \lambda_R \sum_t \left\| \Pi_9 \hat{X}_{t+1}^R \right\|^2 + \lambda_I \sum_t \left\| \Pi_9 \hat{X}_{t+1}^I \right\|^2.$$

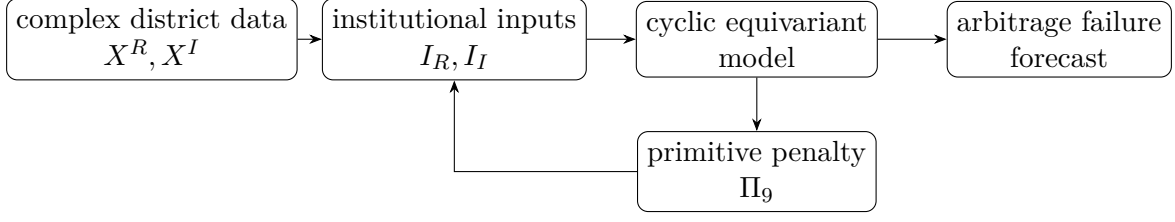


Figure 2: AI/ML architecture for predicting real and imaginary arbitrage failure.

15 Institutional Design Problem

The planner chooses institutions to minimize arbitrage failure:

$$\min_{\alpha^R, \alpha^I} \mathbb{E} \sum_{t=0}^{\infty} \beta^t [D_R(t) + D_I(t) + \lambda_R C_R(t) + \lambda_I C_I(t)]$$

subject to

$$\sum_{(i,j)} c_{ij}^R \alpha_{ij}^R + \sum_{(i,j)} c_{ij}^I \alpha_{ij}^I \leq B,$$

$$0 \leq \alpha_{ij}^R, \alpha_{ij}^I \leq 1.$$

The first-order symbolic condition is

$$\frac{\partial}{\partial \alpha_{ij}^R} \mathbb{E} \sum_t \beta^t [D_R(t) + \lambda_R C_R(t)] = \Lambda c_{ij}^R,$$

$$\frac{\partial}{\partial \alpha_{ij}^I} \mathbb{E} \sum_t \beta^t [D_I(t) + \lambda_I C_I(t)] = \Lambda c_{ij}^I,$$

where Λ is the shadow value of the institutional budget.

Thus the planner invests in real-arbitrage institutions until the marginal reduction in material arbitrage failure equals marginal cost. The planner invests in imaginary-arbitrage institutions until the marginal reduction in expectation arbitrage failure equals marginal cost.

16 Main Characterization

Theorem 3 (Institutional Characterization of Complex Arbitrage). *In the nine-district cyclo-tomic economy, both real and imaginary arbitrage require two institutional systems:*

$$\mathcal{I}^R = \{\text{law, property, payments, settlement, logistics, collateral, clearing}\},$$

and

$$\mathcal{I}^I = \{\text{statistics, disclosure, auditing, communication, derivatives, insurance, credible policy}\}.$$

The first compresses X^R ; the second compresses X^I . If both operate perfectly on prices, rates, and risk premia, the effective sign-state bound falls from

$$3^8 = 6561$$

to

$$3^2 = 9$$

when scarcity remains district-specific, and to

$$1$$

when scarcity is also equalized.

Proof. The raw system has four complex variables and therefore eight ternary sign coordinates. Real arbitrage removes district-specific variation in the real financial coordinates. Imaginary arbitrage removes district-specific variation in the imaginary financial coordinates. If price, rate, and risk-premium signs are all equalized in both components, six coordinates cease to identify districts. The remaining two are the real and imaginary scarcity signs, giving $3^2 = 9$. If scarcity signs are also equalized, no district-specific sign coordinate remains, giving $3^0 = 1$. $\square$

17 Conclusion

The cyclotomic economy with nine districts is not only a mathematical field structure. It is an institutional structure. Its arbitrage behavior depends on whether the economy has institutions capable of compressing both material and anticipatory differences.

Real arbitrage requires institutions of enforceable trade: property rights, contract law, courts, money, settlement, clearing, banking, logistics, and collateral. Imaginary arbitrage requires institutions of enforceable information-contingent trade: statistics, disclosure, auditing, communication, derivatives, insurance, forecasting, and credible public authority.

The raw sign-state bound for complex scarcity, price, rate, and risk premium is

$$6561.$$

But institutions compress this bound. Strong complex financial arbitrage reduces the effective sign capacity to

$$9$$

if scarcity remains locally distinct, and to

$$1$$

if scarcity is also arbitrated and communicated away.

Thus the central conclusion is:

institutions determine how much of the cyclotomic economy remains distinguishable.

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Glossary

Cyclotomic economy

An economy whose district structure is governed by a cyclotomic polynomial.

Ninth cyclotomic polynomial

The polynomial $\Phi_9(x) = x^6 + x^3 + 1$.

Cyclic shift operator

The operator P satisfying $(Px)_i = x_{i-1}$ and $P^9 = I$.

Primitive field

The subspace $\mathcal{H}_9 = \text{Ker } \Phi_9(P)$.

Hidden triad

One of the district triples $\{0, 3, 6\}$, $\{1, 4, 7\}$, or $\{2, 5, 8\}$.

Complex economic quantity

A variable $X = X^R + iX^I$ with material and anticipatory components.

Real component

The material, settled, physical, or realized part of a complex economic variable.

Imaginary component

The expectational, informational, shadow, or anticipatory part of a complex economic variable.

Real arbitrage

Arbitrage that compresses differences in real components across districts.

Imaginary arbitrage

Arbitrage that compresses differences in imaginary components across districts.

Real arbitrage wedge

The material transaction cost τ_{ij}^R preventing real equalization.

Imaginary arbitrage wedge

The information, ambiguity, or credibility cost τ_{ij}^I preventing imaginary equalization.

Componentwise arbitrage

Arbitrage that operates separately on real and imaginary components.

Sign-state bound

The maximum number of districts distinguishable by signs of selected real and imaginary variables.

Effective district capacity

The number $3^{d_{\text{eff}}}$, where d_{eff} is the number of independent ternary sign coordinates remaining after institutional compression.

Institutional compression

The reduction of district-specific sign differences by arbitrage, communication, law, settlement, and information systems.

Primitive geopolitical pressure

A geopolitical pressure vector lying in $\mathcal{H}_9$, invisible to aggregate measures but active in the primitive field.

Arbitrage failure

Persistent dispersion in real or imaginary components beyond institutional wedges.

Cyclotomic institutional loss

A statistical or AI/ML penalty measuring primitive arbitrage failure.

The End